9C Lesson 4

Treating diseases

Year 9 Science · 9C Threat from disease · Student Book pages 44 to 45

Name _____ Class _____ Date _____

Today I can:

☐ I can explain which medicines treat which types of disease, and why antibiotics do not work on viruses.

☐ I can explain how large amounts of human hormones are made quickly using genetically modified bacteria in a fermenter.

Key words 关键词

drug	药物
medicine	药品
antibiotic	抗生素
antiviral	抗病毒药
genetic disorder	遗传病
autoimmune disease	自身免疫病
insulin	胰岛素
transgenic	转基因的
fermenter	发酵罐
sterilise	灭菌
aseptic	无菌的

Command words 指令词

Complete 完成（填写） · Describe 描述 · Draw 画出 · Explain 解释 · Give 给出 · Match 连线（配对） · Tick 打勾

● Core Everyone does this section.

1. Match each scientific word to its definition. Draw a line.

medicines	drugs that prevent viruses replicating
antibiotics	a hormone that makes cells take up glucose from the blood
antivirals	drugs used to treat or prevent disease
vaccines	substances that make us immune to particular diseases
insulin	drugs that kill bacteria

2. Tick one box in each row to show the best treatment for each disease.

Disease	antibiotic	antiviral	add a nutrient	inject a hormone	vaccine (prevents it)
ear infection caused by bacteria					
flu (caused by a virus)					
scurvy (too little vitamin C)					
type 1 diabetes (no insulin made)					
measles, before a child catches it					

3. Complete the text about fermenters using words from the word bank.

Transgenic bacteria are grown in a large container called a _____. Before use it is _____ with steam so that no other _____ can grow inside it. _____ air is

bubbled in so that the bacteria can respire. _____ paddles mix the bacteria, nutrients and oxygen, and a water _____ keeps the temperature steady.

●● Stretch When Core is finished.

S1. Explain why medicines are drugs.

S2. Describe how transgenic bacteria are made to produce human growth hormone.

S3. In type 1 diabetes the body stops making insulin. Explain why this can cause organ damage, and state how the disease is treated.

●●● Challenge Explain in full sentences.

C1. Transgenic bacteria are grown in fermenters.

a) Give the reason why air is bubbled into the fermenter.

b) Describe two ways in which the fermenter keeps the bacteria growing at a high rate.

c) Describe the aseptic precautions taken when preparing the fermenter, and explain why they are needed.

C2. Scientists can also remove genes from an organism. Explain how this can create a genetically modified organism that is not transgenic.

My thinking (any language 可用中文)

Some bacteria are no longer killed by antibiotics that used to work. Suggest why using antibiotics when they are not needed could make this problem worse.

Word bank drug 药物 · medicine 药品 · antibiotic 抗生素 · antiviral 抗病毒药 · vaccine · insulin 胰岛素 · transgenic 转基因的 · fermenter 发酵罐 · sterilised · microorganisms · Filtered · Stirring · jacket

✂ ---------- cut here

Exit ticket 9C Lesson 4: Treating diseases Name _____

1. What is the difference between an antibiotic and an antiviral?

2. What does 'transgenic' mean?

3. Give one reason why a fermenter is sterilised before use.

Today I can: Goal 1 ☐ Goal 2 ☐